CBCS SCHEME - Make-Up Exam

USN | | | | | | | | | | | **BEC401**

Fourth Semester B.E/B.Tech. Degree Examination, June/July 2025
Electromagnetic Theory

Time: 3 hrs.   Max. Marks:100

Note: *1. Answer any FIVE full questions, choosing ONE full question from each module.*
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
1	a.	State vector form of Coulomb's law of force between two point charges and indicate the units of quantities in the equation.	6	L2	CO1
	b.	Q_1 and Q_2 are the point charges located at (0, –4, 3) and (0, 1, 1). If Q_1 is 2nc, find Q_2 such that the force on a test charge at (0, –3, 4) has no Z component.	8	L3	CO1
	c.	Calculate the electric field intensity at a point (3, 4, 5) due to a charge of 5 nc placed at (1, 2, 3).	6	L3	CO1
		OR			
2	a.	Derive an expression for the electric field intensity due to infinite line charge.	8	L2	CO1
	b.	Find $\vec{D}$ in Cartesian co-ordinate system at point P(6, 8, –10) due to: i) a point charge of 40 mc at the origin ii) a uniform line charge of $\rho_L = 40$ μc/m on the Z–axis.	8	L3	CO1
	c.	Define electric flux and flux density.	4	L1	CO1
		Module – 2			
3	a.	State and prove Gauss law as applied to an electric field.	8	L3	CO2
	b.	The flux density $D = \frac{r}{3}\vec{a}_r$ nc / m^2 is in the free space i) Find $\vec{E}$ at r = 0.2m ii) Find the total electric flux leaving the sphere of r = 0.2 m iii)Find the total charge within the sphere of r = 0.3m.	8	L3	CO2
	c.	Find the divergence of $\vec{A}$ at $P\left(5, \frac{\pi}{2}, 1\right)$ where $A = r\ Z\sin\phi\,\vec{a}_r + 3rZ^2\cos\phi\,\vec{a}_\phi$.	4	L3	CO2
		OR			
4	a.	State and prove Gauss divergence theorem.	8	L3	CO2
	b.	If the potential field V is $V = 100(x^2 - y^2)$. Find $\vec{E}$, V at a point (2, –1, 3) and the equation representing the locus of all points having a potential of 300 V.	4	L3	CO2
	c.	Derive continuity of current equation.	8	L2	CO2
		Module – 3			
5	a.	Using Biot-Savart's law, determine the magnetic field intensity at a pint due to infinite long straight conductor.	7	L3	CO3
	b.	Verify the potential field given below satisfies the Laplace's equation : $V = 2x^2 - 3y^2 + z^2$.	5	L3	CO3
	c.	Derive Laplace and Poisson's equations and write Laplace equation in all 3 co-ordinate systems.	8	L2	CO3

		OR			
6	a.	State and explain Amperes circuital law.	8	L2	CO3
	b.	Given that the general vector $\vec{A}$ is $\vec{H} = 2.5\vec{a}_\theta + \vec{a}_\phi$ in spherical co-ordinates. Find the curl of H at $(2, \pi/6, 0)$.	6	L3	CO3
	c.	Given that the vector magnetic potential $\vec{A} = x^2\,\vec{a}_x + 2yz\,\vec{a}_y + (-x)^2\,\vec{a}_z$. Find the magnetic flux density.	6	L3	CO3
		Module – 4			
7	a.	Derive the expression for the force between two differential current elements.	6	L2	CO4
	b.	A point charge of Q = –1.2c has velocity $\vec{V} = (5\vec{a}_x + 2\vec{a}_y - 3\vec{a}_z)$m/sec. Find the magnitude of the force exerted on the charge if : i) $\vec{E} = -18\vec{a}_x + 5\vec{a}_y - 10\vec{a}_z$ V/m ii) $\vec{B} = -4\vec{a}_x + 4\vec{a}_y + 3\vec{a}_z$ T iii) Both are present simultaneously.	9	L3	CO4
	c.	A conductor 6 m long lies along Z-direction with a current of 2A in $\vec{a}_z$ direction. Find the force experienced by conductor if $\vec{B} = 0.08\vec{a}_x$ T.	5	L3	CO4
		OR			
8	a.	Write a note on : i) Magnetization ii) Permeability iii) Forces on magnetic materials.	6	L1	CO4
	b.	If $\vec{B} = 0.05x\,\vec{a}_y$ T in a material for which $x_m = 2.5$. Find : i) μ_r ii) μ iii) $\vec{H}$ iv) $\vec{M}$ v) $\vec{J}$.	8	L3	CO4
	c.	Discuss on magnetic boundary conditions.	6	L2	CO4
		Module – 5			
9	a.	List Maxwell's equations for steady and time varying field in : i) Point form ii) Integral form.	6	L2	CO5
	b.	State and explain Faraday's law of electromagnetic induction.	6	L2	CO5
	c.	If the magnetic field $\vec{H} = [3x\ \cos\beta + 6y\sin\alpha]\vec{a}_z$. Find current density $\vec{J}$ if fields are invariant with time.	8	L3	CO5
		OR			
10	a.	Obtain solution of the wave equation for a uniform plane wave in free space.	8	L1	CO5
	b.	State and prove Poynting theorem.	8	L3	CO5
	c.	Wet marshy soil is characterized by $\sigma = 10^{-2}$ s/m, $\epsilon_r = 15$ and $\mu_r = 1$. Show that at 60 Hz, it can be considered as good conductor. Hence at 60 Hz calculate : i) Skin depth ii) Intrinsic impedance iii) Propagation constant.	4	L3	CO3

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CBCS SCHEME

USN 2JI22EC098 　　　　　　　　　　　　　　　　　　　　　　　　BEC401

Fourth Semester B.E./B.Tech. Degree Examination, June/July 2024
Electromagnetics Theory

Time: 3 hrs. 　　　　　　　　　　　　　　　　　　　　　　　　Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	State and explain spherical coordinate system in detail.	5	L2	CO1
	b.	Four point charges each of 10 μC are placed in free space at the points (1, 0, 0), (-1, 0, 0), (0, 1, 0) and (0, -1, 0) m respectively. Determine the force on a point charge of 30 μC located at a point (0, 0, 1) m.	8	L3	CO1
	c.	Show that electric field intensity at a point, due to 'n' number of point charges, is given by, $E = \frac{1}{4\pi\varepsilon_0}\sum_{i=1}^{n}\frac{Q_i}{R_i^2}a_{R_i}$ V/m	7	L3	CO1
		OR			
Q.2	a.	Define electric field intensity. Derive the expression for electric field intensity due to infinite line charge.	9	L1	CO1
	b.	Given the two points $A(\rho = 4.4, \phi = -115^\circ, Z = 2)$ and B(x = -3.1, y = 2.6, z = -3), find (i) The rectangular coordinate of point A (ii) The cylindrical coordinate of point B (iii) The distance between A and B.	5	L3	CO1
	c.	Find E at P(1, 5, 2) m in free space if a point charge of 6 μC is located at (0, 0, 1), the uniform line charge density $\rho_L = 180$ nC/m along x axis.	6	L3	CO1
		Module – 2			
Q.3	a.	State and prove Gauss's law for point charge.	6	L3	CO2
	b.	Calculate the divergence of D at the point specified if, (i) $D = (2xyz - y^2)a_x + (x^2z - 2xy)a_y + x^2y a_z$ C/m² at P_A (2, 3, -1) (ii) $D = 2\rho Z^2 \sin^2\phi a_\rho + \rho Z^2 \sin 2\phi a_\phi + 2\rho^2 Z \sin^2\phi a_z$ C/m² at $P_B(\rho = 2, \phi = 110^\circ, Z = -1)$ (iii) $D = 2r\sin\theta\cos\phi a_r + r\cos\theta\cos\phi a_\theta - r\sin\phi a_\phi$ C/m² at $P_C(r = 1.5, \theta = 30^\circ, \phi = 50^\circ)$	9	L3	CO2
	c.	Find electric field intensity at the point A(1, 2, -1) given the potential $V = 3x^2y + 2y^2z + 3xyz$	5	L3	CO2
		OR			
Q.4	a.	Evaluate both sides of divergence theorem if $D = \frac{5r^2}{4}a_r$ C/m² in spherical co-ordinate for the volume enclosed by r = 4 m and $\theta = \frac{\pi}{4}$ radians.	8	L3	CO2

	b.	Calculate the work done in moving a charge 4C from B(1, 0, 0) to A(0, 2, 0) along the path $y = 2 - zx$, $z = 0$ in the field (i) $E = 5a_x$ V/m (ii) $E = 5xa_x$ V/m (iii) $E = 5xa_x + 5ya_y$ V/m	6	L3	CO2
	c.	Electrical potential at an arbitrary point in free space is given as, $V = 2(x+1)^2(y+2)^2(z+3)^2$ volt at a point P(2, -1, 4). Find (i) V (ii) E (iii) $\lvert E \rvert$ (iv) $\lvert D \rvert$ (v) ρ_v	6	L3	CO2
		Module – 3			
Q.5	a.	Evaluate the expression for capacitance of two uniformly charged parallel planes of infinite extent.	8	L2	CO3
	b.	Determine whether or not the potential equations satisfies Laplaces equation : (i) $V = 2x^2 - 4y^2 + z^2$ (ii) $V = \phi\cos\phi + z$ (iii) $V = r^2\cos\phi + 0$	5	L3	CO3
	c.	An assembly of two concentric spherical shell is considered. The inner spherical shell is at a distance of 0.1 m and is at a potential of 0 volts. The outer spherical shell is at a distance of 0.2 m and at a potential of 100 V. The medium between them is a free space. Find E and D using spherical co-ordinate system.	7	L3	CO3
		OR			
Q.6	a.	State and explain Biot-Savarts law applicable to magnetic field.	6	L2	CO3
	b.	Evaluate both sides of the stokes theorem for the field, $H = 6xya_x - 3y^2a_y$ A/m and the rectangular path around the region, $2 \le x \le 5$, $-1 \le y \le 1$, $Z = 0$. Let the positive direction of ds be a_z.	8	L3	CO3
	c.	Let $A = (3y - z)a_x + 2xza_y$ wb/m in a certain region of free space. (i) Show that $\nabla . A = 0$ (ii) At P(2, -1, 3) find A, B, H and J.	6	L3	CO3
		Module – 4			
Q.7	a.	Obtain the expression for magnetic force between differential current elements.	6	L1	CO4
	b.	The point charge $Q = 18nC$ has a velocity of 5×10^6 m/s in the direction $a_V = 0.60a_x + 0.75a_y + 0.30a_z$. Calculate the magnitude of force exerted on the charge by the field. (i) $B = -3a_x + 4a_y + 6a_z$ mT (ii) $E = -3a_x + 4a_y + 6a_z$ KV/m	6	L3	CO4
	c.	The magnetization in a magnetic material for which $\chi_m = 8$ is given in a certain region as $150\,Z^2a_x$ A/m. At Z = 4 cm, find the magnitude of, i) J_T ii) J iii) J_b.	8	L3	CO4
		OR			
Q.8	a.	Obtain the magnetic boundary conditions at interface between two different magnetic material.	8	L2	CO4
	b.	Two differential current elements $I_1dl_1 = 10^{-4}a_z$ Am at $P_1(1, 0, 0)$ and $I_2dl_2 = 3\times10^{-6}(-0.5a_x + 0.4a_y + 0.3a_z)$ Am at $P_2(2, 2, 2)$ are located in free space. Find the vector force exerted on, (i) I_2dl_2 by I_1dl_1 (ii) I_1dl_1 by I_2dl_2	6	L3	CO4

	c.	The interface between two different regions is normal to one of three Cartesian axes. If $B_1 = \mu_0(43.5a_x + 24.0a_y)$ and $B_2 = \mu_0(22a_x + 24a_y)$. What is the ratio $\frac{\tan\theta_1}{\tan\theta_2}$?	6	L3	C
		Module – 5			
Q.9	a.	For the given medium $\varepsilon = 4\times10^{-9}$ F/m and $\sigma = 0$. Find K so that the following pair of fields satisfies Maxwell's equation, $E = (20y - Kt)a_x$ V/m, $H = (y+2\times10^{6}t)a_z$ A/m.	6	L3	C
	b.	Within a certain region $\varepsilon = 10^{-11}$ F/m and $\mu = 10^{-5}$ H/m, If $B = 2\times10^{-4}\cos10^{5}t\sin10^{-3}y$ T ; (i) Find E (ii) Find total magnetic flux passing through the surface $x = 0$, $0 < y < 40$ m, $0 < z < 2$m at $t = 1$ µsec.	8	L3	C
	c.	State and explain pointing theorem.	6	L2	C
		OR			
Q.10	a.	Derive the modified Ampere's law by Maxwells for time varying fields.	5	L2	C
	b.	Show that the intrinsic impedance of the perfect dielectric $\eta = \frac{\lvert E\rvert}{\lvert H\rvert} = \sqrt{\frac{\mu}{\varepsilon}}$ and show that its value in free space is 377 Ω	7	L2	C
	c.	A plane electromagnetic wave having a frequency of 10 MHz has an average pointing vector of 1 W/m^2. If medium is lossless with relative permeability of 2 and relative permittivity of 3 find (i) The velocity of propagation. (ii) Wavelength. (iii) Impedance of the medium (iv) rms electric field.	8	L3	C

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USN

BEC401

Fourth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Electromagnetic Theory

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

			M	L	C
		Module – 1			
Q.1	a.	State and explain Coulomb's law of force between two point charges in vector form.	8	L1	CO1
	b.	Define Electric field intensity. Derive the expression for the electric field intensity at a point due to infinite line charges (Uniformly charged wire).	8	L2	CO1
	c.	Two very small conducting spheres, each of mass 1 × 10-4 kg are suspended at common point by very thin filaments of length 0.2m. A charge Q Coulomb is placed on each sphere. The electric force of repulsion separates the spheres and an equilibrium is reached when the suspending filaments make an angle of 10°. Assuming $e_r = 1$, $g = 9.8$N/kg and negligible mass for the filaments, find Q.	4	L3	CO1
		OR			
Q.2	a.	Define Point charge and using Coulomb's Law, derive expression for electric field intensity due to a point charge.	8	L2	CO1
	b.	Let a point Q1 = 25nc be located at A(4, -2, 7) and a charge Q2 = 60nc be at B(-3, 4, -2). Find $\bar{E}$ at C(1, 2, 3). Also find direction of the electric field. Given $\varepsilon_0 = 8.854 \times 10^{-12}$ F/m.	8	L3	CO1
	c.	Two point charges of $+3 \times 10^{-9}$ C and -2×10^{-9} C are spaced two meter apart. Determine the electric field at a point which is one meter from each of the two point charges.	4	L3	CO1
		Module – 2			
Q.3	a.	State and prove Gauss Divergence theorem or divergence theorem.	8	L2	CO2
	b.	A point charge , Q = 30nc is located at the origin in Cartesian coordinates. Find the electric flux density and electric field intensity at (1, 3, -4)m.	8	L3	CO2
	c.	Derive an equation for equation of continuity (continuity of current).	4	L3	CO2
		OR			
Q.4	a.	State and prove Gauss law.	8	L2	CO2
	b.	Given that the potential field is $V = 2x^2y - 5z$. Find the potential , electric field intensity and volume charge density at point P(-4, 3, 6).	8	L3	CO2
	c.	State Gauss law in point form. Hence derive Maxwell's first equation.	4	L3	CO2

		Module – 3			
Q.5	a.	Starting from gauss law, derive Poisson's and Laplace equation. Hence define Laplace equation in all three coordinate systems.	4	L2	CO3
	b.	State and prove Stoke's theorem.	8	L2	CO3
	c.	Find the potential and volume charge density at P(0.5 , 1.5 , 1)m in free space. Given the potential field as under. i) $V = 2x^2 - y^2 - z^2$ volt ii) $V = 6\, r\, \phi\, z$ volt.	8	L3	CO3
		OR			
Q.6	a.	State and prove Biot – Savart's law.	4	L1	CO3
	b.	State and prove Ampere's circuital law.	8	L1	CO3
	c.	The magnetic field intensity is given in a certain region of space as : $\vec{H} = \left(\dfrac{x+2y}{z^2}\right)\hat{a}_y + \dfrac{2}{z}\hat{a}_z$ A/m. i) Find $\nabla \times \vec{H}$ ii) Find $\vec{J}$ iii) Use $\vec{J}$ to find total current passing through the surface , $Z = 4$, $1 < x < 2$, $3 < y < 5$ in the $\hat{a}_z$ direction.	8	L3	CO3
		Module – 4			
Q.7	a.	Define current element. Derive an equation for force on a differential current element in a magnetic field.	8	L2	CO4
	b.	A point charge Q = 18nc has a velocity of 5×10^6 m/s in the direction $\vec{a} = 0.6\,\hat{a}_x + 0.75\,\hat{a}_y + 0.3\,\hat{a}_z$. Calculate the magnitude of the force exerted on the charge by the field $\vec{B} = -3\,\hat{a}_x + 4\,\hat{a}_y + 6\,\hat{a}_z$ mT.	8	L3	CO4
	c.	Calculate the force on a straight conductor of length 0.3m carrying a current 5A in the Z – direction where the magnetic field is $\vec{B} = 3.5 \times 10^{-3}\,(a\hat{x} - a\hat{y})$ Tesla. ($a\hat{x}$ and $a\hat{y}$ are unit vectors).	4	L3	CO4
		OR			
Q.8	a.	Derive magnetic boundary condition for i) Tangential component of magnetic field. ii) Normal component of magnetic field.	8	L2	CO4
	b.	A conductor 4m long lies along the Y – axis with a current of 10A in the $a\hat{y}$ direction. Find the force on the conductor if the field in the region is $\vec{B} = 0.05\, a\hat{x}$ tesla.	8	L3	CO4
	c.	Find the magnetic field intensity inside a magnetic material for following conditions : $M = 100$A/m and $\mu = 1.5 \times 10^{-5}$ H/m $B = 200\mu T$, X_m (Magnetic susceptibility = 15).	4	L3	CO4

		Module – 5			
Q.9	a.	Derive Integral and point form of Faraday's law.	8	L2	CO5
	b.	Given $\bar{E} = E_m \sin(wt - \beta z)\, a\hat{y}$ in free space. Calculate $\bar{D}$, $\vec{B}$ and $\bar{H}$.	8	L3	CO5
	c.	A copper disc 40cm diameter is rotated at 3000 r.p.m on a horizontal axis perpendicular to and through the centre of disc axis, lying in magnetic meridian. Two brushes make contact with the disc at diametrically opposite points on the edge. If horizontal component of earth's field is 0.02 mT, find the induced e.m.f between brushes.	4	L3	CO5
		OR			
Q.10	a.	State and derive Poynting's theorem for uniform plane waves.	8	L2	CO5
	b.	Derive general wave equation in electric and magnetic fields.	8	L2	CO5
	c.	For silver, the conductivity is $\sigma = 3.0 \times 10^6$ s/m. At what frequency will depth of penetration be 1mm?	4	L3	CO5

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Model Question Paper-1 with effect from 2022-23 (CBCS Scheme)

USN

Fourth Semester B.E. Degree Examination

Subject: ELECTROMAGNETIC THEORY

TIME: 03 Hours **Max. Marks: 100**

Note: 01. Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE**.
02.
03.

		Module -1	***Bloom's Taxonomy Level**	**Marks**
Q.01	a	State and explain the Cylindrical coordinate systems in detail	L1	05
	b	Show that electric field intensity at a point, due to 'n' number of point charges, is given by $\vec{E} = \frac{1}{4\pi\epsilon_o}\sum_{i=1}^{n}\frac{Q_i}{R_i^2}\hat{a}_{R_i}$ V / m.	L3	07
	c	Two point charges, $Q_1 = 30\ \mu C$ and $Q_2 = -100\ \mu C$, are located at (2,0,5) and (-1,0,-2)m respectively. Find (i) force on Q_1(ii) force on Q_2 (iii) the magnitude of forces (iv) directions of forces	L3	08
		OR		
Q.02	a	State Coulomb's law of force between any two point charges and indicate the units of the quantities involved.	L1	06
	b	Show that the electric field intensity at any point due to an infinite sheet of charge is independent of the distance to the point from the sheet.	L3	08
	c	Two point charges, Q1 and Q2 are located at (1, 2, 0)m and (2, 0, 0)m respectively. Find the relation between the charges, Q1 and Q2 such that the total force on a unit positive charge at (-1, 1, 0) have (i). no x-component, (ii). no y-component.	L3	08
		Module-2		
Q. 03	a	State and prove Gauss's law for point charge.	L1	06
	b	A point charge, Q = 90 μC is located at the origin and there are surface charge distributions $-8\ \mu C/m^2$ at r = 1 m and $4.5\ \mu C/m^2$ at r = 2 m. Find $\vec{D}$ everywhere.	L3	08
	c	Calculate the divergence of D at the point specified if (i) $\vec{D} = (1/z^2)[10xyz\ \hat{a}_x + 5x^2z\ \hat{a}_y + (2z^3-5x^2y)\ \hat{a}_z]$ at P(−2, 3, 5) (ii) $\vec{D} = 5z^2\hat{a}_r + 10\ r\ z\ \hat{a}_z$ at $P(3, -45^0, 5)$ (iii) $\vec{D} = 2\ r \sin\theta \sin\Phi\ \hat{a}_r + r\cos\theta\sin\Phi\ \hat{a}_\theta + r\cos\Phi\ \hat{a}_\Phi$ at $P(3, 45^0, 45^0)$	L3	08
		OR		
Q.04	a	State & prove Divergence theorem.	L1	07
	b	Obtain an expression for electric field intensity due to an infinite line charge along z- axis having a uniform charge of ρ_L C/m using Gauss's law	L3	06
	c	Given $\vec{D} = 0.3\ r^2\ a_r$ nC/m^2 in free space, (a) find $\vec{E}$ at P(2, 25°, 90°) (b) find the total charge within the sphere, r = 3. (c) Find the total electric flux leaving the sphere, r = 4, m.	L3	07

	b	I. Write a short note on: Skin effect in conductors. II. What do you mean by depth of penetration?	L1	05
	c	With respect to wave propagation in good conductors, describe what the skin effect is and derive an expression for the depth of penetration. If $\sigma = 58 \times 10^6$ mhos/m at frequency 10 MHz, determine the depth of penetration.	L3	08

*Bloom's Taxonomy Level: Indicate as L1, L2, L3, L4, etc. It is also desirable to indicate the COs and POs to be attained by every bit of questions.

Module-3				
Q. 05	a	Derive expression for potential and capacitance between the planes at z=0 and z=d if the potential v = V_1 and v = V_2 respectively using Laplace's equation.	L3	06
	b	If potential of $V = x^2yz + Ay^3z$ Volts, i) find A so that the Laplace's equation is satisfied. ii) With that value A, determine the electric field at a point p whose coordinates are (2, 1, -1).	L3	08
	c	There exists a potential of V = -2.5 volts on a conductor at0.02 m and V = 15 volts at r = 0.35 m. Determine E and D by solving the Laplace's equation in spherical coordinates representing the potential system.	L3	08
OR				
Q. 06	a	Derivation of Ampere's Circuital Law in point form using Strokes theorem.	L3	06
	b	Derive Poisson's and Laplace equations and writ Laplace equation in cylindrical and polar coordinates.	L3	06
	c	Long concentric and right conducting cylinders in free space, at r = 5 mm and r = 25 mm in cylindrical coordinates have voltages of zero and V_o respectively. If the electric field intensity, $\vec{E}$= -8.28×10$^3\hat{a}_r$ at r = 15 mm, find V_o and the charge density on the outer conductor by using Laplace's equation.	L3	08
Module-4				
Q. 07	a	State and Explain the force between differential Current Elements.	L1	05
	b	Find the force per meter length between two long parallel wires separated by 10cm in air and carrying current of 100A in opposite direction state the nature of force between wires	L3	07
	c	Find magnetization in a magnetic material where, (i)$\mu = 1.8X10^{-5}\left(\frac{H}{m}\right)$ $and\ M = 120\ \left(\frac{H}{m}\right)$. (ii) $\mu_r = 22$, there are 8.3$X10^{28}$ atoms/m^3 and each atom has dipole moment of 4.5X10^{-27} (A.M^2) and (iii) B=300(μT) and Xm=15.	L3	08
OR				
Q. 08	a	Write short notes Magnetic Boundary Conditions.	L1	05
	b	Derive the equations for Magnetic circuits with suitable diagram.	L3	07
	c	A conductor 4m long lies along the y-axis with a current of 10A in the $\overrightarrow{a_y}$ direction. Find the force on the conductor if the field in the region is (in region is) $\vec{B} = 0.005\overrightarrow{a_x}$ Tesla.	L3	08
Module-5				
Q. 09	a	What is a uniform wave? Explain its propagation in free space with necessary equations	L1	05
	b	Starting from Maxwell's equations, derive the wave equation for sinusoidal waves in good dielectric medium.	L3	07
	c	A 9.375 GHz uniform plane wave is propagating in polyethylene (ϵ_r = 2.26). If the amplitude of the electric field is 500 V/m and the material is assumed to be lossless, find (i) Phase constant (ii) Wavelength (iii) Velocity of propagation (iv) Intrinsic impedance (v) magnetic field intensity	L3	08
OR				
Q. 10	a	Show that the uniform plane wave is transverse in nature.	L3	07

Model Question Paper-1 with effect from 2022-23 (CBCS Scheme)

USN

Fourth Semester B.E. Degree Examination

Subject: ELECTROMAGNETIC THEORY

TIME: 03 Hours **Max. Marks: 100**

Note: 01. Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE**.
02.
03.

		Module -1	*Bloom's Taxonomy Level	Marks
Q.01	a	State and explain vector form of Coulomb's law	L1	06
	b	Infinite line charges of 5nC/m along the (positive and negative) x and y axes in free space. Find E at $P_A(0,0,4)$	L3	04
	c	Four 10nC positive charges are located in z=0 plane at the corners of a square 8cm on a side. A fifth 10nC charge is located at a point 8cm distant from other charges. Calculate the magnitude of total force on the fifth charge for $\varepsilon = \varepsilon_0$	L3	10
		OR		
Q.02	a	Given the two points, C(-3,2,1) and D(r=5, θ=20°, Φ=-70°), find a. the spherical coordinates of C b. the rectangular coordinates of D	L1	05
	b	Derive electric field intensity due to line charge density	L3	08
	c	Find Electric field intensity at origin if the following charge distributions are present in free space a. Point charge 12nC at P(2,0,6) b. uniform line charge density 3nC/m at x=-2 and y=3 c. uniform surface charge density $0.2nC/m^2$ at x=2	L3	07
		Module-2		
Q. 03	a	State and explain Gauss's law and write point form and integral form of Gauss law.	L1	06
	b	Evaluate both sides of divergence theorem if $\mathbf{D}=\frac{5r^2}{4}\mathbf{a_r}$ C/m^2 in spherical coordinate for the volume enclosed between r=1m and r=2m	L3	06
	c	Find the volume charge density at the points indicated if i. $D = 4rz\sin\Phi\, a_r + 2rz\cos\Phi\, a_\Phi + 2r^2\sin\Phi\, a_z$ C/m^2 at $P_A(1, \pi/2, 2)$ ii. $D = \sin\Theta\cos\Phi\, a_r + \cos\Theta\sin\Phi\, a_\Theta - \sin\Phi\, a_\Phi$ at $P_B(2, \pi/3, \pi/6)$	L3	08
		OR		
Q.04	a	Derive the equation of continuity	L1	06
	b	Derive the expression for the work done in moving a point charge within an electric field.	L3	06
	c	Find the work done in moving a 5 μC charge from origin to P(2,-1,4) through $E= 2xyz\, a_x + x^2z\, a_y + x^2y\, a_z$ V/m; via the path i. Straight line segments (0, 0, 0) to (2, 0, 0) to (2, -1, 0) to (2, -1, 4) ii. Straight line x = - 2 y; z = 2x	L3	08
		Module-3		
Q. 05	a	Derive Poisson's and Laplace's Equation	L3	06
	b	Given the potential field $V = (Ar^4 + Br^{-4})\sin(4\Phi)$. Show that potential field satisfies Laplace's equation. Also find A and B such that V =100 V and $\lvert E\rvert$ = 500 V/m at P(1,22.5°, 2)	L3	08
	c	Evaluate expression for capacitance of spherical shell using Laplaces equation	L3	08
		OR		

Q. 06	a	Derivation of Ampere's Circuital Law in point form using Strokes theorem.	L3	06
	b	If the magnetic field intensity in a region is $H = (3y-2)\,a_z + 2x\,a_y$ A/m, find the current density at origin. And Calculate the value of J if $H=\frac{1}{\sin\theta}\,a_\theta$ A/m at P(2,30°, 20°)	L3	08
	c	Long concentric and right conducting cylinders in free space, at r = 5 mm and r = 25 mm in cylindrical coordinates have voltages of zero and V_o respectively. If the electric field intensity, $\vec{E}= -8.28\times10^3\hat{a}_r$ at r = 15 mm, find V_o and the charge density on the outer conductor by using Laplace's equation.	L3	08
		Module-4		
Q. 07	a	State and Explain the force between differential Current Elements.	L1	05
	b	Find the force per meter length between two long parallel wires separated by 10cm in air and carrying current of 100A in opposite direction state the nature of force between wires	L3	07
	c	Let the permittivity be 5 µH/m in a region A where x<0 and 20µH/m in region B, where x>0, If there is a surface current density $K = 150\,a_y - 200\,a_z$ A/m at x=0 and $H_A = 300\,a_x - 400\,a_y + 500\,a_z$ A/m. Find i. $\lvert H_{tA}\rvert$ ii. $\lvert H_{NA}\rvert$ iii. $\lvert H_{tB}\rvert$ iv. $\lvert H_{NB}\rvert$	L3	08
		OR		
Q. 08	a	Determine the boundary conditions for the magnetic field at the interface between two different magnetic materials	L1	08
	b	Derive the equations for Magnetic circuits with suitable diagram.	L3	07
	c	The point charge Q= 18nC has a velocity of 5×10^6m/s in the direction of $a_v = 0.6\,a_x + 0.75\,a_y + 0.3\,a_z$ calculate the magnitude of the force exerted on the charge by the field i. $B = -3\,a_x + 4a_y + 6a_z$ mT ii. $E = -3\,a_x + 4a_y + 6a_z$ KV/m	L3	05
		Module-5		
Q. 09	a	What is a uniform wave? Explain its propagation in free space with necessary equations	L1	05
	b	What is inconsistency of Ampere's law with continuity equation? Derive the modified Ampere's law by Maxwell for time varying fields	L3	07
	c	A 9.375 GHz uniform plane wave is propagating in polyethylene ($\epsilon_r = 2.26$). If the amplitude of the electric field is 500 V/m and the material is assumed to be lossless, find (i) Phase constant (ii) Wavelength (iii) Velocity of propagation (iv) Intrinsic impedance (v) magnetic field intensity	L3	08
		OR		
Q. 10	a	Show that the uniform plane wave is transverse in nature.	L3	07
	b	State and explain Poynting Theorem .	L1	05
	c	With respect to wave propagation in good conductors, describe what the skin effect is and derive an expression for the depth of penetration. If $\sigma = 58 \times 10^6$ mhos/m at frequency 10 MHz, determine the depth of penetration.	L3	08

*Bloom's Taxonomy Level: Indicate as L1, L2, L3, L4, etc. It is also desirable to indicate the COs and POs to be attained by every bit of questions.